# MULTI-SITE SOLAR PV PORTFOLIO (37.8 kW)

**Date: 21-Sep-2026**

Portfolio: Karnataka Institutional Portfolio (2 Sites, 3 Buildings) | Location: Shivamogga, Karnataka | Grid-Tied LT Dossier

| PLANT CAPACITY | ANNUAL YIELD | CAPITAL COST | PAYBACK PERIOD |
|---|---|---|---|
| 400W Mono | Specific Yield | 12-Item BOM | 25Y Lifecycle |
| **37.75 kW** | **59,854 kWh** | **Rs. 1,776,014** | **4Y 8M** |
| 32.00 kWp Installed | CUF 18.1% \| PR 75% | Rs. 17.76 Lakhs | Profit: Rs. 175.86L |

## [SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

| | |
|---|---|
| **Company / Customer Name** | Karnataka Institutional Portfolio |
| **Location / Address** | Shivamogga, Karnataka |
| **Contact Email Address** | vidya@example.com |
| **Design Path Selected** | Rooftop Area Analysis |
| **System Target Capacity** | 37.75 kW (2 Sites Portfolio) |
| **Total Installed Array Rating** | 32.00 kWp |
| **Solar PV Module Rating** | 400W Mono PERC / TOPCon Half-Cut |
| **Total Module Count** | 80 Nos. |

## [METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

| | |
|---|---|
| **Total Rooftop Area** | 4375.00 Sq.Feet (406.45 sq.m) |
| **Utilized Usable Rooftop Area** | 3775 Sq.Feet (350.71 sq.m) |
| **First Year Annual Generation** | 59,854.15 kWh |
| **25-Year Cumulative Generation** | 1,329,831.35 kWh |
| **Estimated Payback Period** | 4 Years & 8 Months |
| **Post-Payback Profitability (25 Yrs)** | Rs. 17,585,706.00 (INR 175.86 Lakhs) |
| **Alternative FD 6.5% Benchmark (25 Yrs)** | Rs. 8,574,060.62 (INR 85.74 Lakhs) |

## ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 1090 Tonnes CO2 | Tree Equivalence: 1744 Mature Trees | Green Energy: 100% Clean LT Power

*Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.*

## [PORTFOLIO] Multi-Site Portfolio Allocation & Regulatory Schedule

Aggregated LT solar portfolio overview, site-wise capacities, and statutory compliance

| TOTAL SITES | TOTAL CAPACITY | TURNKEY CAPITAL | ANNUAL YIELD |
|---|---|---|---|
| PORTFOLIO | AGGREGATE | 12-ITEM BOM | CLEAN POWER |
| **2 Sites** | **37.75 kW** | **Rs. 1,776,014** | **59,854 kWh** |
| 3 Buildings Total | 80 Panels (400W) | Rs. 17.76 Lakhs | 1st Year Aggregate |

## [ALLOCATION] Site-by-Site Allocation & Engineering Capacity Matrix

Independent LT net-meter installations conforming to 20 kW per-connection statutory ceiling

| Sl. | Site Name & Location | Bldgs | Gross Area | Capacity | Inverter | 1st Yr Yield | Turnkey Cost |
|---|---|---|---|---|---|---|---|
| 01 | Vidyanagar Campus (Shivamo.. | 2 Nos. | 2,175 sq.ft | 18.48 kW | 19 kW 3-Ph | 29,301 kWh | Rs. 870,188 |
| 02 | Technology Park (Bengaluru.. | 1 Nos. | 2,200 sq.ft | 19.27 kW | 20 kW 3-Ph | 30,553 kWh | Rs. 905,826 |
| | **TOTAL PORTFOLIO SUMMARY** | | **4,375 sq.ft** | **37.75 kW** | **39 kW Total** | **59,854 kWh** | **Rs. 1,776,014** |

### KERC & ESCOM LOW TENSION (LT) 20 kW REGULATORY COMPLIANCE MANDATE

Under Karnataka Electricity Regulatory Commission (KERC) Rooftop Solar PV regulations and ESCOM interconnect guidelines, individual Low Tension (LT) net-metered service connections are subject to a statutory capacity ceiling of 20.0 kW per consumer installation / meter account. Interconnections above 20 kW necessitate High Tension (HT 11 kV) infrastructure, dedicated transformer bays, and elevated statutory approvals.

To maintain full regulatory compliance while meeting multi-site commercial / institutional power demand, this portfolio is architected into discrete, independent LT installations. Each individual site operates with its own dedicated bidirectional net-meter, solar inverter, ACDB protection board, and safety switchgear, while engineering specifications, equipment warranties, and turnkey financials are consolidated within this unified dossier.

### LT PORTFOLIO STATUTORY & GRID INTERCONNECTION STANDARDS

**Metering Architecture:** Independent Bi-Directional TOD Net-Meters per site (1 - 20 kW LT Class)

**Anti-Islanding Protection:** Mandatory automatic grid disconnection upon utility outage (IEC 62116 / IEEE 1547 compliant)

**Switchgear & Isolation:** Dedicated IP65 ACDB with Type-II AC/DC SPDs & 10 kA Breaking MCBs per connection

**DISCOM Permitting:** Separate online DISCOM portal filings with aggregated turnkey EPC execution and common monitoring

## [ARRAY] Vidyanagar Campus -- Building 1: Layout & Array Sizing

Site: Vidyanagar Campus (Shivamogga, Karnataka) | Sub-Array: 13.44 kW | Site Inverter: 19 kW

| | |
|---|---|
| **Building / Array Identification** | Vidyanagar Campus -- Main Academic Block (Rooftop Array) |
| **Gross Rooftop Area** | 1575.00 Sq.Feet (146.32 sq.m) |
| **Utilized Effective Rooftop Area** | 1344 Sq.Feet (124.86 sq.m) |
| **Building Plant Target Capacity** | 13.44 kW |
| **Solar PV Module Rating** | 400 W Mono PERC / TOPCon Half-Cut |
| **Number of Solar Modules** | 34 Nos. |
| **String Sizing Architecture** | 18 - 20 Panels per MPPT String (Optimal Voltage Band) |
| **Selected Site Inverter Rating** | 19 kW, 415V 3-Ph MPPT |

## [INVERTER] Selected Grid-Tied Inverter Specifications (19 kW)

Core inverter rating, MPPT topology, and operating parameters

| | |
|---|---|
| **Selected Inverter Rating** | 19 kW Grid-Tied Solar Inverter |
| **AC Output Voltage & Grid Phase** | 415V, 3-Phase, 50 Hz AC |
| **MPPT Operating Topology** | Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency) |
| **Enclosure & Grid Protection** | IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116) |
| **Inverter to ACDB Cable** | 3.5Cx35 Sq.mm |
| **DC Array Isolator Switch** | 80A, MCB Type, 4P 415V |

## [MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

| | |
|---|---|
| **Incomer Overcurrent Protection** | Thermal-Magnetic MCB / MCCB with Manual Isolation |
| **Breaker Current Rating** | 40 Amps |
| **Rated Inverter Operating Current** | 26.43 Amps |
| **Trip Curve & Breaking Capacity** | Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1) |
| **Surge Protection Device (SPD)** | Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling |

## [CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

| | |
|---|---|
| **Solar PV DC Array Cable** | 4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618) |
| **Inverter AC Output Cable** | 3.5Cx35 Sq.mm (Cu / XLPE / Armoured) |
| **ACDB to Utility Grid Cable** | 3.5Cx16 Sq.mm (Al / XLPE / Armoured) |
| **Earth Grounding Conductor** | 16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits |
| **Design Standard & Voltage Drop** | Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52 |

## [ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

| | |
|---|---|
| **ACDB Enclosure Specification** | IP65 Weatherproof Polycarbonate / Sheet Steel |
| **Internal Busbar Rating** | 50A, 415V, 3PN Aluminium |
| **Incomer & Outgoing Terminations** | Cable Glands with Compression Lugs & Dual Earthing |
| **Interconnection Distance** | 25.0 Meters (ACDB to Utility Net-Meter) |

## [ARRAY] Vidyanagar Campus -- Building 2: Layout & Array Sizing

Site: Vidyanagar Campus (Shivamogga, Karnataka) | Sub-Array: 5.04 kW | Site Inverter: 19 kW

| | |
|---|---|
| **Building / Array Identification** | Vidyanagar Campus -- Science Laboratory (Rooftop Array) |
| **Gross Rooftop Area** | 600.00 Sq.Feet (55.74 sq.m) |
| **Utilized Effective Rooftop Area** | 504 Sq.Feet (46.82 sq.m) |
| **Building Plant Target Capacity** | 5.04 kW |
| **Solar PV Module Rating** | 550 W Mono PERC / TOPCon Half-Cut |
| **Number of Solar Modules** | 10 Nos. |
| **String Sizing Architecture** | 18 - 20 Panels per MPPT String (Optimal Voltage Band) |
| **Selected Site Inverter Rating** | 19 kW, 415V 3-Ph MPPT |

## [INVERTER] Selected Grid-Tied Inverter Specifications (19 kW)

Core inverter rating, MPPT topology, and operating parameters

| | |
|---|---|
| **Selected Inverter Rating** | 19 kW Grid-Tied Solar Inverter |
| **AC Output Voltage & Grid Phase** | 415V, 3-Phase, 50 Hz AC |
| **MPPT Operating Topology** | Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency) |
| **Enclosure & Grid Protection** | IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116) |
| **Inverter to ACDB Cable** | 3.5Cx35 Sq.mm |
| **DC Array Isolator Switch** | 80A, MCB Type, 4P 415V |

## [MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

| | |
|---|---|
| **Incomer Overcurrent Protection** | Thermal-Magnetic MCB / MCCB with Manual Isolation |
| **Breaker Current Rating** | 40 Amps |
| **Rated Inverter Operating Current** | 26.43 Amps |
| **Trip Curve & Breaking Capacity** | Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1) |
| **Surge Protection Device (SPD)** | Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling |

## [CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

| | |
|---|---|
| **Solar PV DC Array Cable** | 4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618) |
| **Inverter AC Output Cable** | 3.5Cx35 Sq.mm (Cu / XLPE / Armoured) |
| **ACDB to Utility Grid Cable** | 3.5Cx16 Sq.mm (Al / XLPE / Armoured) |
| **Earth Grounding Conductor** | 16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits |
| **Design Standard & Voltage Drop** | Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52 |

## [ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

| | |
|---|---|
| **ACDB Enclosure Specification** | IP65 Weatherproof Polycarbonate / Sheet Steel |
| **Internal Busbar Rating** | 50A, 415V, 3PN Aluminium |
| **Incomer & Outgoing Terminations** | Cable Glands with Compression Lugs & Dual Earthing |
| **Interconnection Distance** | 25.0 Meters (ACDB to Utility Net-Meter) |

## [ARRAY] Technology Park -- Building 1: Layout & Array Sizing

Site: Technology Park (Bengaluru, Karnataka) | Sub-Array: 19.27 kW | Site Inverter: 20 kW

| | |
|---|---|
| **Building / Array Identification** | Technology Park -- Tower A Innovation Hub (Rooftop Array) |
| **Gross Rooftop Area** | 2200.00 Sq.Feet (204.39 sq.m) |
| **Utilized Effective Rooftop Area** | 1927 Sq.Feet (179.02 sq.m) |
| **Building Plant Target Capacity** | 19.27 kW |
| **Solar PV Module Rating** | 550 W Mono PERC / TOPCon Half-Cut |
| **Number of Solar Modules** | 36 Nos. |
| **String Sizing Architecture** | 18 - 20 Panels per MPPT String (Optimal Voltage Band) |
| **Selected Site Inverter Rating** | 20 kW, 415V 3-Ph MPPT |

## [INVERTER] Selected Grid-Tied Inverter Specifications (20 kW)

Core inverter rating, MPPT topology, and operating parameters

| | |
|---|---|
| **Selected Inverter Rating** | 20 kW Grid-Tied Solar Inverter |
| **AC Output Voltage & Grid Phase** | 415V, 3-Phase, 50 Hz AC |
| **MPPT Operating Topology** | Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency) |
| **Enclosure & Grid Protection** | IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116) |
| **Inverter to ACDB Cable** | 3.5Cx35 Sq.mm |
| **DC Array Isolator Switch** | 80A, MCB Type, 4P 415V |

## [MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

| | |
|---|---|
| **Incomer Overcurrent Protection** | Thermal-Magnetic MCB / MCCB with Manual Isolation |
| **Breaker Current Rating** | 40 Amps |
| **Rated Inverter Operating Current** | 27.82 Amps |
| **Trip Curve & Breaking Capacity** | Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1) |
| **Surge Protection Device (SPD)** | Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling |

## [CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

| | |
|---|---|
| **Solar PV DC Array Cable** | 4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618) |
| **Inverter AC Output Cable** | 3.5Cx35 Sq.mm (Cu / XLPE / Armoured) |
| **ACDB to Utility Grid Cable** | 3.5Cx16 Sq.mm (Al / XLPE / Armoured) |
| **Earth Grounding Conductor** | 16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits |
| **Design Standard & Voltage Drop** | Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52 |

## [ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

| | |
|---|---|
| **ACDB Enclosure Specification** | IP65 Weatherproof Polycarbonate / Sheet Steel |
| **Internal Busbar Rating** | 50A, 415V, 3PN Aluminium |
| **Incomer & Outgoing Terminations** | Cable Glands with Compression Lugs & Dual Earthing |
| **Interconnection Distance** | 15.0 Meters (ACDB to Utility Net-Meter) |

## [SITE BOM] Site 1: Vidyanagar Campus -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Vidyanagar Campus (18.48 kW)

| Sl. | Component / Material Description | Engineering Specification | Qty | Amount (INR) |
|---|---|---|---|---|
| 01 | Solar PV modules | Main Academic Block: 400W (34 Nos.) + Science Laboratory: 550W (10 Nos.) | 44 Nos. | Rs. 347,282 |
| 02 | On-grid inverter | 19 kW three-phase (Main Academic Block: 14kW + Science Laboratory: 6kW) | 2 Nos. | Rs. 200,556 |
| 03 | Rooftop mounting structure | HDGI, module-compatible | 2 Roof Set(s) | Rs. 90,000 |
| 04 | DC solar cable | 6 sq.mm copper, PV-rated | 190 m | Rs. 16,150 |
| 05 | AC cable | 25 sq.mm copper | 100 m | Rs. 35,000 |
| 06 | MC4 connectors | PV-rated connector pairs | 44 pairs | Rs. 13,200 |
| 07 | DCDB | Isolator + fuse + Type-II DC SPD | 2 No(s). | Rs. 16,000 |
| 08 | ACDB | Main Academic Block: 25A MCB + Science Laboratory: 25A MCB + Type-II AC SPD | 2 Set(s) | Rs. 16,000 |
| 09 | Earthing kit | Earth electrodes & accessories (IS 3043 - 3 Pits) | 3 Nos. | Rs. 7,500 |
| 10 | Lightning arrester | Standard rooftop PV type | 1 No. | Rs. 6,500 |
| 11 | Installation accessories | Lugs, ferrules, clamps, conduit, glands, etc. | 1 Lot | Rs. 32,000 |
| 12 | Installation & commissioning | Labour and commissioning | 1 Job | Rs. 90,000 |
| | | **TOTAL PLANT-SIDE CAPITAL COST (Vidyanagar Campus)** | | **Rs. 870,187.56** |

## [UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Vidyanagar Campus)

Official DISCOM statutory filing fees and net-metering charges for this connection

| | |
|---|---|
| **Application & Registration Fee** | Rs. 1,000 + GST (DISCOM portal filing) |
| **Facilitation & Technical Processing Fee** | Rs. 2,000 + GST (Feasibility clearance) |
| **PPA e-Sign Stamp Duty Charge** | Rs. 11.8 (Electronic stamping) |
| **Bidirectional Utility Net-Meter** | Rs. 12,000 (DISCOM meter testing/procurement) |
| **Refundable Security Deposit** | Rs. 8,000 (Refundable utility escrow) |

**TOTAL ESTIMATED CAPITAL OUTLAY FOR VIDYANAGAR CAMPUS**

Turnkey Plant BOM: Rs. 870,188 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 893,199 (Rs. 8.93 Lakhs)

## [SITE BOM] Site 2: Technology Park -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Technology Park (19.27 kW)

| Sl. | Component / Material Description | Engineering Specification | Qty | Amount (INR) |
|---|---|---|---|---|
| 01 | Solar PV modules | 550 W mono PERC / TOPCon | 36 Nos. | Rs. 360,000 |
| 02 | On-grid inverter | 20 kW, three-phase | 1 No. | Rs. 210,526 |
| 03 | Rooftop mounting structure | HDGI, module-compatible | 1 Set | Rs. 95,000 |
| 04 | DC solar cable | 6 sq.mm copper, PV-rated | 200 m | Rs. 17,000 |
| 05 | AC cable | 25 sq.mm copper | 100 m | Rs. 35,000 |
| 06 | MC4 connectors | PV-rated connector pairs | 36 pairs | Rs. 10,800 |
| 07 | DCDB | Isolator + fuse + Type-II DC SPD | 1 No. | Rs. 17,000 |
| 08 | ACDB | 40A MCB + 63A RCCB + Type-II AC SPD | 1 No. | Rs. 17,000 |
| 09 | Earthing kit | Earth electrodes & accessories (IS 3043 - 3 Pits) | 3 Nos. | Rs. 7,500 |
| 10 | Lightning arrester | Standard rooftop PV type | 1 No. | Rs. 7,000 |
| 11 | Installation accessories | Lugs, ferrules, clamps, conduit, glands, etc. | 1 Lot | Rs. 34,000 |
| 12 | Installation & commissioning | Labour and commissioning | 1 Job | Rs. 95,000 |
| | | **TOTAL PLANT-SIDE CAPITAL COST (Technology Park)** | | **Rs. 905,826.32** |

## [UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Technology Park)

Official DISCOM statutory filing fees and net-metering charges for this connection

| | |
|---|---|
| **Application & Registration Fee** | Rs. 1,000 + GST (DISCOM portal filing) |
| **Facilitation & Technical Processing Fee** | Rs. 2,000 + GST (Feasibility clearance) |
| **PPA e-Sign Stamp Duty Charge** | Rs. 11.8 (Electronic stamping) |
| **Bidirectional Utility Net-Meter** | Rs. 12,000 (DISCOM meter testing/procurement) |
| **Refundable Security Deposit** | Rs. 8,000 (Refundable utility escrow) |

**TOTAL ESTIMATED CAPITAL OUTLAY FOR TECHNOLOGY PARK**

Turnkey Plant BOM: Rs. 905,826 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 928,838 (Rs. 9.29 Lakhs)

## [PORTFOLIO TOTAL] Consolidated Portfolio Capital Cost & Grand Total Investment

Consolidated turnkey capital outlay across all portfolio sites and DISCOM budgets

| TOTAL CAPACITY | TURNKEY BOM | ESCOM BUDGET | TOTAL CAPITAL |
|---|---|---|---|
| AGGREGATE | 12-ITEM BOM | UTILITY | GRAND TOTAL |
| **37.75 kW** | **Rs. 17.76L** | **Rs. 0.46L** | **Rs. 18.22L** |
| 2 Sites Sized | Rs. 1,776,014 | Rs. 46,024 | Rs. 1,822,037 |

## [CAPEX MATRIX] Portfolio Site-by-Site Capital Investment Matrix

Comprehensive turnkey capital expenditure required for all sites

| Sl. | Site Name & Location | Buildings | Capacity | Plant BOM Cost | ESCOM Budget | Total Outlay |
|---|---|---|---|---|---|---|
| 01 | Vidyanagar Campus | 2 Nos. | 18.48 kW | Rs. 870,188 | Rs. 23,012 | Rs. 893,199 |
| 02 | Technology Park | 1 Nos. | 19.27 kW | Rs. 905,826 | Rs. 23,012 | Rs. 928,838 |
| **GRAND TOTAL CAPITAL REQUIRED (ALL SITES)** | | | **37.75 kW** | **Rs. 1,776,014** | **Rs. 46,024** | **Rs. 1,822,037** |

** Marked installation(s) capped at statutory 20.0 kW Low Tension (LT) ceiling in accordance with KERC regulations.*

### PORTFOLIO PROCUREMENT, EPC EXECUTION & DISCOM GRID CONNECTIVITY

All electrical installations conform to CEA Technical Standards for Distributed Generation and State Grid Code. Turnkey procurement bundles Tier-1 TOPCon modules and high-efficiency inverters under consolidated bulk pricing, while each site operates under its dedicated independent DISCOM net-meter account (RR number). This avoids High Tension (HT 11 kV) transformer infrastructure and maintains complete statutory compliance.

## [FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

| CAPITAL INVESTMENT (CAPEX) | PAYBACK PERIOD (RETURN) | 25Y NET SAVINGS (PROFIT) | FD 6.5% BENCHMARK (BENCHMARK) |
|---|---|---|---|
| **Rs. 1,776,014** | **4Y 8M** | **Rs. 175.86L** | **Rs. 85.74L** |
| 12-Item BOM Basis | Break-Even Year | Rs. 17,585,706 | Alternative Asset |

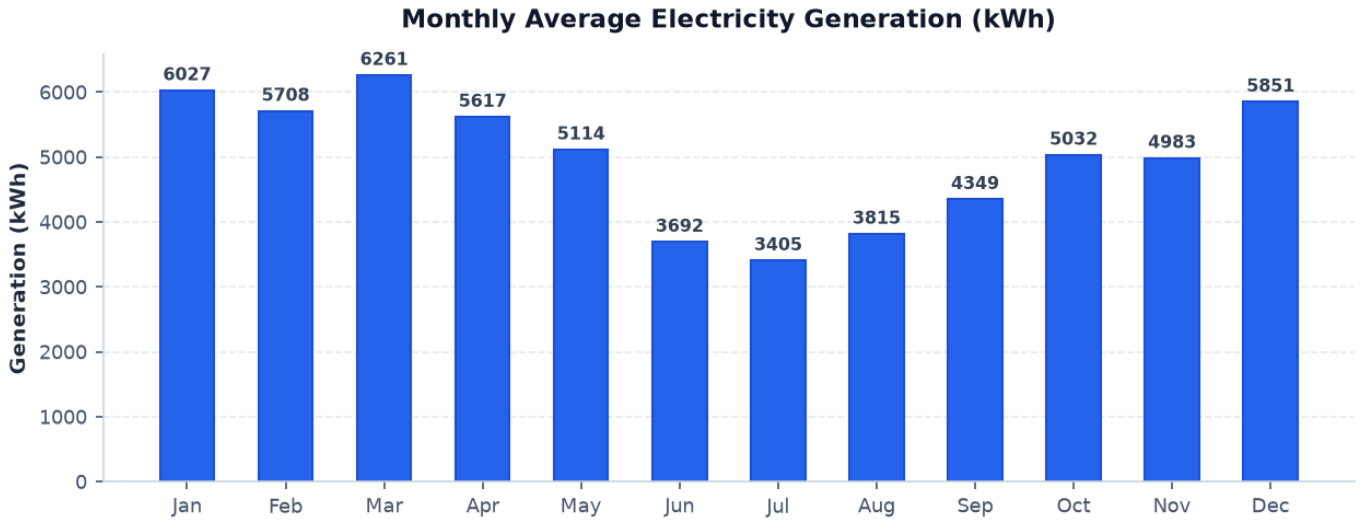
| Yr | Gen (kWh) | Tariff (Rs/u) | Gross Rev (Rs) | AMC (Rs) | Net Annual (Rs) | Cumulative Net (Rs) |
|---|---|---|---|---|---|---|
| 01 | 59,854 | Rs. 8.05 | 481,826 | 22,650 | 459,176 | 459,176 |
| 02 | 59,256 | Rs. 8.45 | 500,710 | 22,650 | 478,060 | 937,236 |
| 03 | 58,663 | Rs. 8.87 | 520,341 | 22,650 | 497,691 | 1,434,927 |
| **04 *** | **58,076** | **Rs. 9.32** | **541,272** | **22,650** | **518,622** | **1,953,549** |
| 05 | 57,496 | Rs. 9.78 | 562,308 | 22,650 | 539,658 | 2,493,207 |
| 06 | 56,921 | Rs. 10.27 | 584,576 | 22,650 | 561,926 | 3,055,132 |
| 07 | 56,351 | Rs. 10.78 | 607,469 | 22,650 | 584,819 | 3,639,952 |
| 08 | 55,788 | Rs. 11.32 | 631,520 | 22,650 | 608,870 | 4,248,821 |
| 09 | 55,230 | Rs. 11.89 | 656,686 | 22,650 | 634,036 | 4,882,857 |
| 10 | 54,678 | Rs. 12.48 | 682,379 | 22,650 | 659,729 | 5,542,586 |
| 11 | 54,131 | Rs. 13.11 | 709,658 | 22,650 | 687,008 | 6,229,594 |
| 12 | 53,590 | Rs. 13.76 | 737,394 | 22,650 | 714,744 | 6,944,338 |
| 13 | 53,054 | Rs. 14.45 | 766,628 | 22,650 | 743,978 | 7,688,316 |
| 14 | 52,523 | Rs. 15.18 | 797,303 | 22,650 | 774,653 | 8,462,969 |
| 15 | 51,998 | Rs. 15.93 | 828,329 | 22,650 | 805,679 | 9,268,648 |
| 16 | 51,478 | Rs. 16.74 | 861,743 | 22,650 | 839,093 | 10,107,741 |
| 17 | 50,963 | Rs. 17.57 | 895,425 | 22,650 | 872,775 | 10,980,516 |
| 18 | 50,454 | Rs. 18.45 | 930,870 | 22,650 | 908,220 | 11,888,736 |
| 19 | 49,949 | Rs. 19.37 | 967,514 | 22,650 | 944,864 | 12,833,600 |
| 20 | 49,450 | Rs. 20.34 | 1,005,805 | 22,650 | 983,155 | 13,816,755 |
| 21 | 48,955 | Rs. 21.36 | 1,045,681 | 22,650 | 1,023,031 | 14,839,787 |
| 22 | 48,466 | Rs. 22.43 | 1,087,083 | 22,650 | 1,064,433 | 15,904,219 |
| 23 | 47,981 | Rs. 23.55 | 1,129,951 | 22,650 | 1,107,301 | 17,011,520 |
| 24 | 47,501 | Rs. 24.73 | 1,174,702 | 22,650 | 1,152,052 | 18,163,572 |
| 25 | 47,026 | Rs. 25.96 | 1,220,797 | 22,650 | 1,198,147 | 19,361,720 |

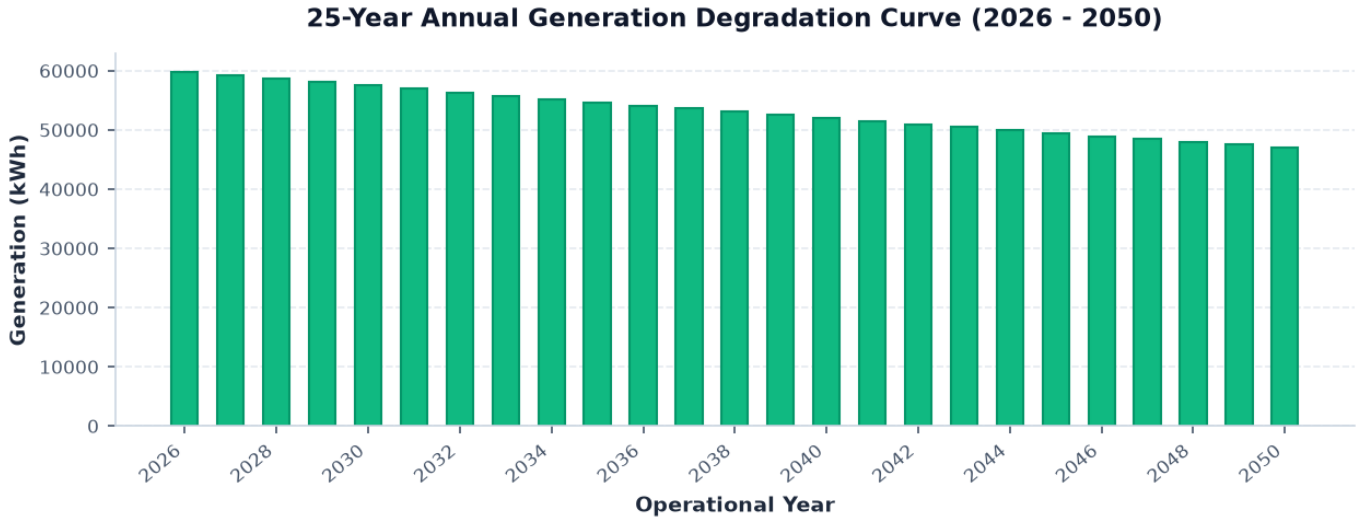
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

## [ANALYTICS] Generation Analytics & Performance Curves

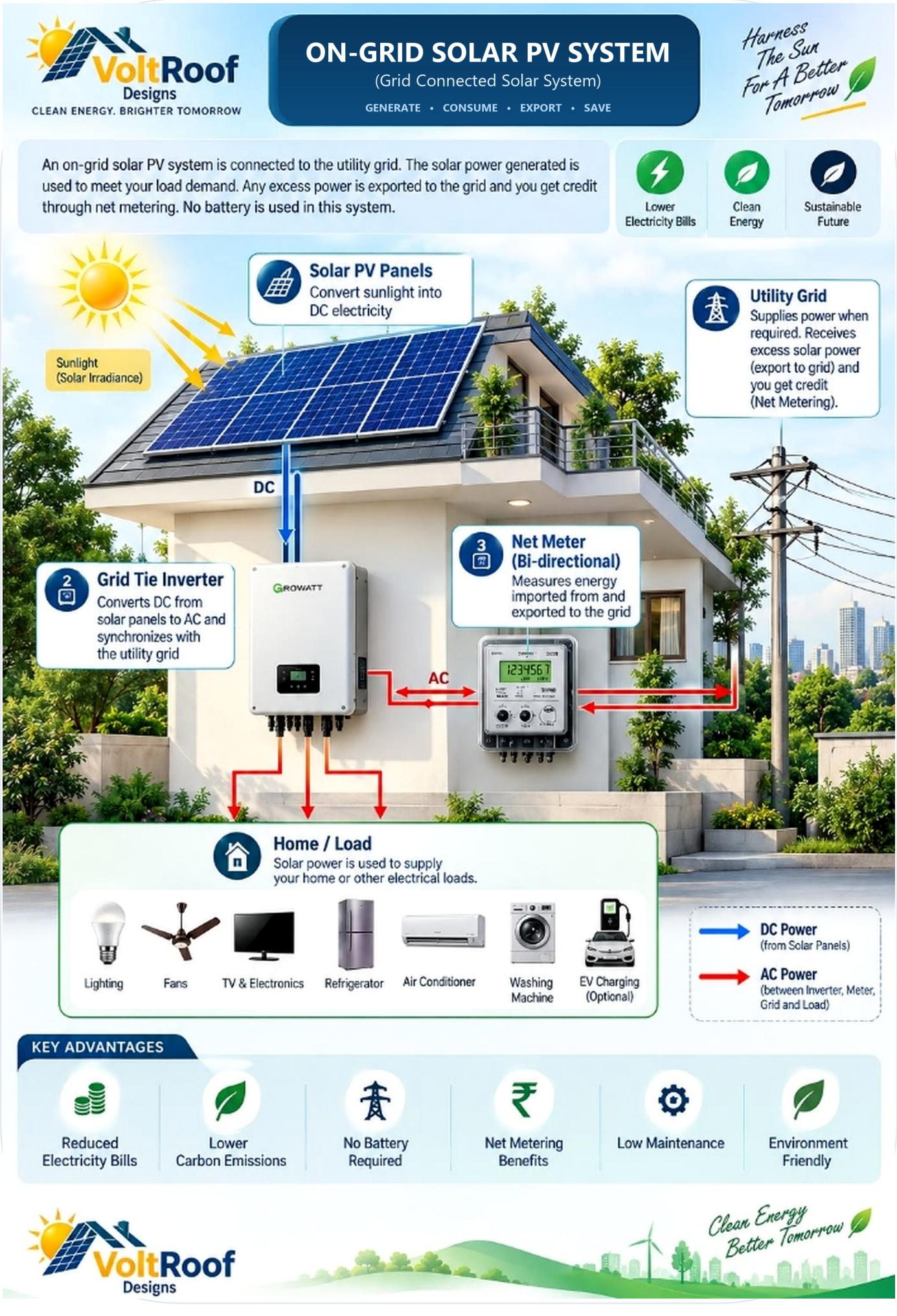
Monthly solar insolation yield and 25-year degradation profile

| SPECIFIC YIELD (YIELD) | PERFORMANCE RATIO (PR) | CAPACITY UTIL. (CUF) | DEGRADATION RATE (DEGRADATION) |
|---|---|---|---|
| **1586 kWh/kWp** | **75.0%** | **18.1%** | **1.0% / year** |
| kWh/kWp/Year | IEC 61724 Standard | Annual Plant Factor | Tier 1 Linear Warranty |




*Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.*


VoltRoof Designs
CLEAN ENERGY. BRIGHTER TOMORROW
ON-GRID SOLAR PV SYSTEM
(Grid Connected Solar System)
GENERATE • CONSUME • EXPORT • SAVE
Harness The Sun For A Better Tomorrow
An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.
Lower Electricity Bills
Clean Energy
Sustainable Future
Solar PV Panels
Convert sunlight into DC electricity
Sunlight (Solar Irradiance)
Utility Grid
Supplies power when required. Receives excess solar power (export to grid) and you get credit (Net Metering).
DC
2 Grid Tie Inverter
Converts DC from solar panels to AC and synchronizes with the utility grid
GROWATT
3 Net Meter (Bi-directional)
Measures energy imported from and exported to the grid
AC
Home / Load
Solar power is used to supply your home or other electrical loads.
Lighting
Fans
TV & Electronics
Refrigerator
Air Conditioner
Washing Machine
EV Charging (Optional)
DC Power (from Solar Panels)
AC Power (between Inverter, Meter, Grid and Load)
KEY ADVANTAGES
Reduced Electricity Bills
Lower Carbon Emissions
No Battery Required
Net Metering Benefits
Low Maintenance
Environment Friendly
VoltRoof Designs
Clean Energy Better Tomorrow